



MSC THESIS DEFENCE

The Impact of Ireland's Proposed LDES Procurement Mechanism on Battery Storage Deployment: A System Dynamics Simulation

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Before we proceed, here are some key facts about how the power grid and battery projects operate.

01

Supply and demand must balance in real time

Electricity cannot be stored on the grid, so power systems rely on dispatchable resources (e.g. BESS, hydro, natural gas) alongside variable renewables (e.g. solar and wind).

02

Merit order effect determines electricity price

Electricity price is set by the most expensive generator needed to meet demand.

03

Curtailed energy is potential renewable electricity that goes unused

When renewable output exceeds the grid's capacity to accommodate it, the excess generation is curtailed.

04

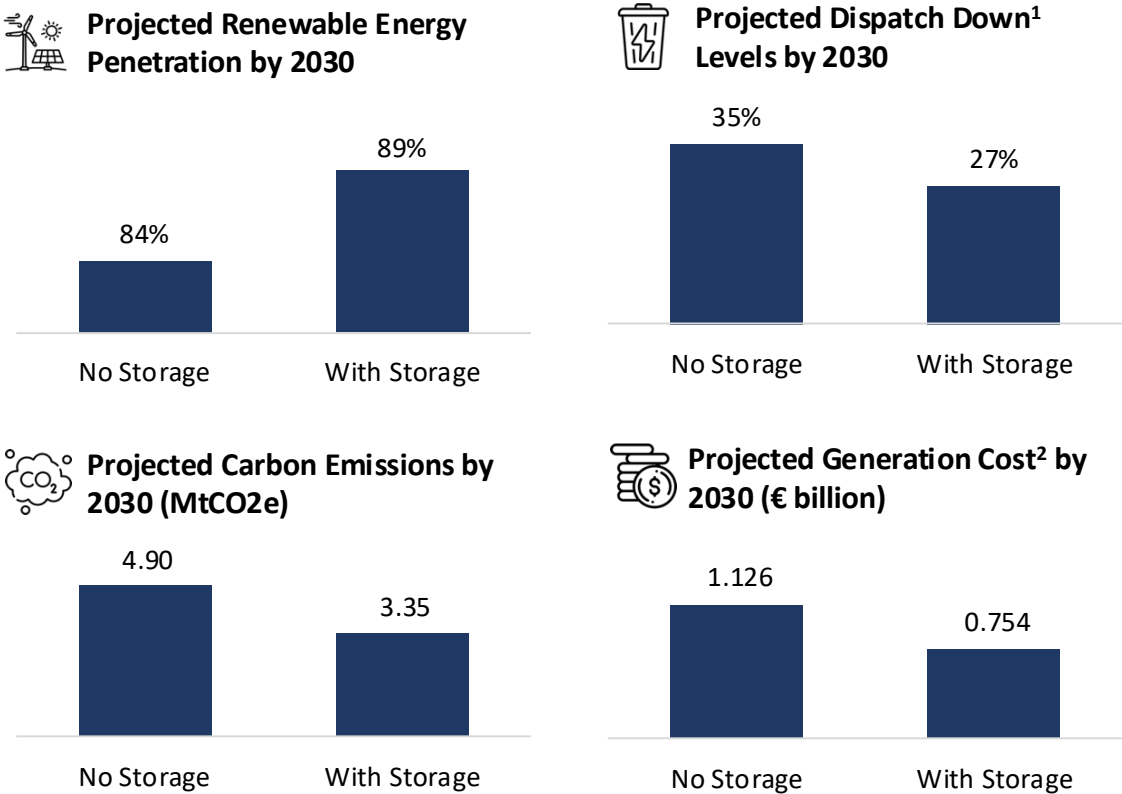
Batteries earn revenue via electricity arbitrage

They store electricity when prices are low and sell it back to the grid when prices are high.

PROBLEM STATEMENT

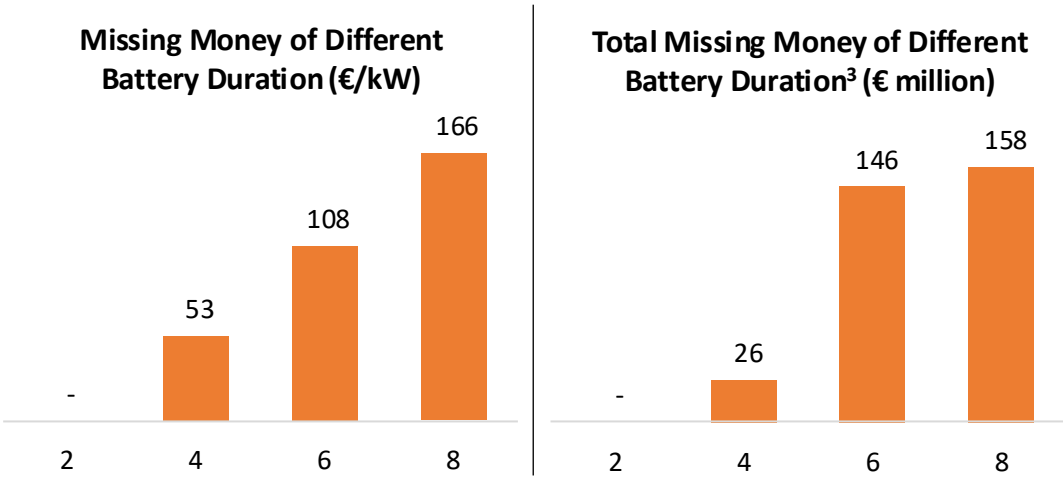
Ireland recognises that energy storage is essential to achieving its climate targets, but market failure prevents sufficient deployment without policy intervention.

THE NEED



EirGrid’s SOEF v1.1 projection shows that strategically increasing long-duration storage could increase renewable energy penetration, reduce wastage, carbon emissions and generation cost.

THE PROBLEM



- “Missing money” is the shortfall between expected market revenues and the investment required to make LDES financially viable.
- As storage duration increases, so does the missing money gap.
- For the desired 3,850 MW of 2-8-hour storage targeted in the SOEF v1.1 scenario, the total missing money equates to approx. €330 million.
- Left alone, the private market would under-invest in something that the power system needs

To address this, EirGrid published a consultation paper in October 2025 proposing a procurement mechanism for Long-Duration Energy Storage (LDES)

01

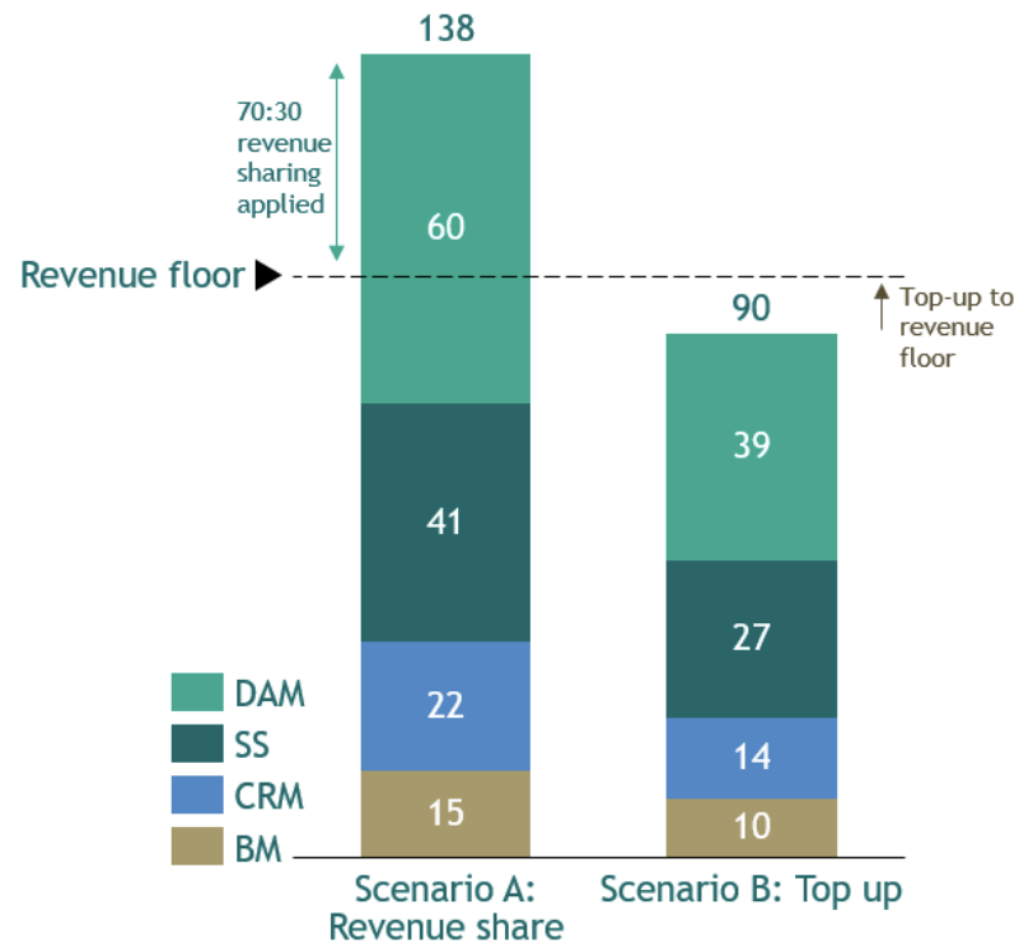
Governance via Operational Envelope

- EirGrid sets an operational envelope specifying when and how much each battery can charge and discharge.

02

Remuneration via Revenue Floor and Sharing

- Developers bid a Revenue Floor linked to their Levelised Cost of Storage (LCOS).
- If revenues fall below the Revenue Floor, EirGrid covers the shortfall.
- If revenues exceed the Revenue Floor, excess revenues are shared 70:30 (developer:TSO).



Illustrative revenue-floor top-up mechanism (EirGrid, 2025)

Hence, this research aims to answer the following:

How much LDES investment, in terms of BESS capacity, would EirGrid's proposed LDES procurement mechanism bring in from 2027 to 2039?

- **Sub-RQ1:** To what extent would a Revenue Floor stimulate the deployment of LDES?
- **Sub-RQ2:** What is the projected impact of contracted LDES assets on renewable curtailment volumes?
- **Sub-RQ3:** What is the policy cost of the procurement mechanism across the modelled scenarios?

KEY METRICS

Operational BESS Capacity

Arbitrage Spread

Curtailment Avoided

Cost of Policy

I followed Sterman's four-phase System Dynamics process to define, structure, build, and test the model

SYSTEM DYNAMIC PHASES	
Phase	Activities
Problem Articulation	Problem: Can the Revenue Floor + Operational Envelope mechanism close the funding gap for viable investment? System boundary: • BESS revenue from I-SEM day-ahead only • Demand held constant • Fleet charges/discharges at the same time daily • 2027–2039 (12-year horizon)
Dynamic Hypothesis Formulation	• Causal Loop Diagram (CLD) • Stock-and-Flow Diagram (SFD)
Simulation Model Formulation	• Developed stock & auxiliary equations • Parameterise constants • Developed table functions (Sterman's nine-step procedure) • Defined policy scenarios
Testing and Evaluation	• Sensitivity analysis on the judgement-based table function (investment-response lookup) • Tested for behaviour-mode, policy, and numerical sensitivity

THE 3-POLICY SCENARIO TESTED			
Policy Scenario	Revenue Floor Factor	Envelope Restriction	Description
Balanced	0.90	0.3	Balanced compromise between battery developer security and TSO control.
Maximum Cost Recovery	1.0	0.7	Maximum financial underwriting with maximum TSO control.
Minimum Cost Recovery	0.7	0.2	Minimal revenue protection with high operational freedom for the developer.

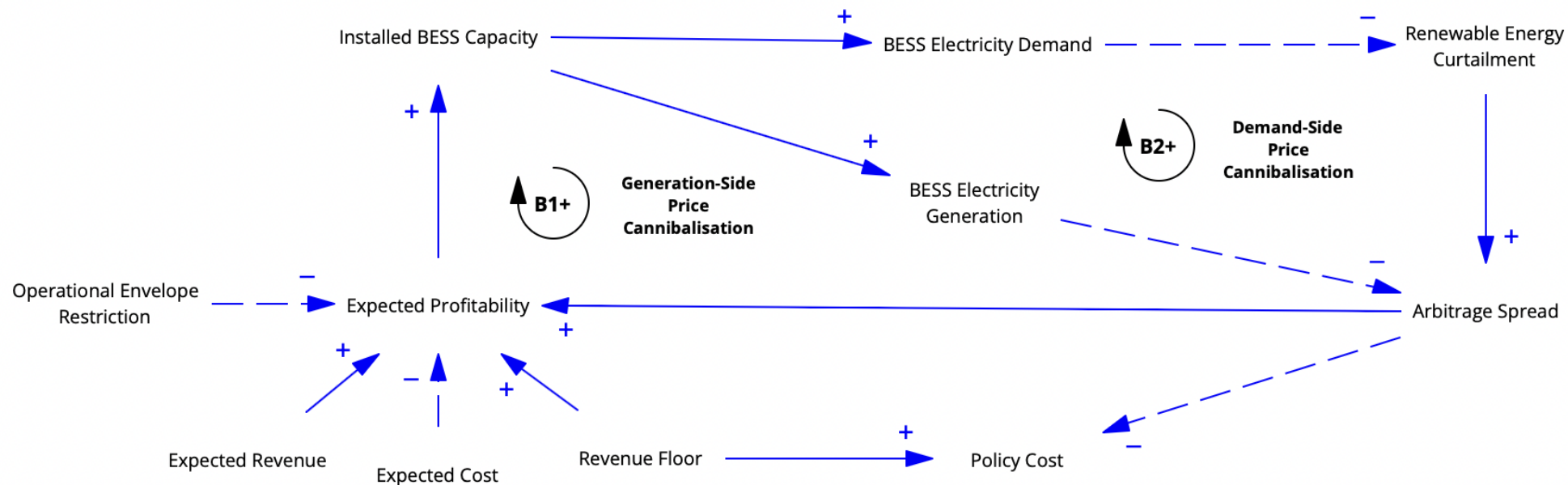
Revenue Floor Factor

0.90 means that EirGrid guarantees the revenue equivalent to 90% of the project's LCOS

Envelope Restriction

0.30 means that 30% of the project's unconstrained revenue are foregone due to EirGrid's Operational Envelope

The system is governed by two balancing feedback loops: increasing BESS deployment reduces electricity selling prices while increasing charging price



B1 — Generation-side cannibalisation

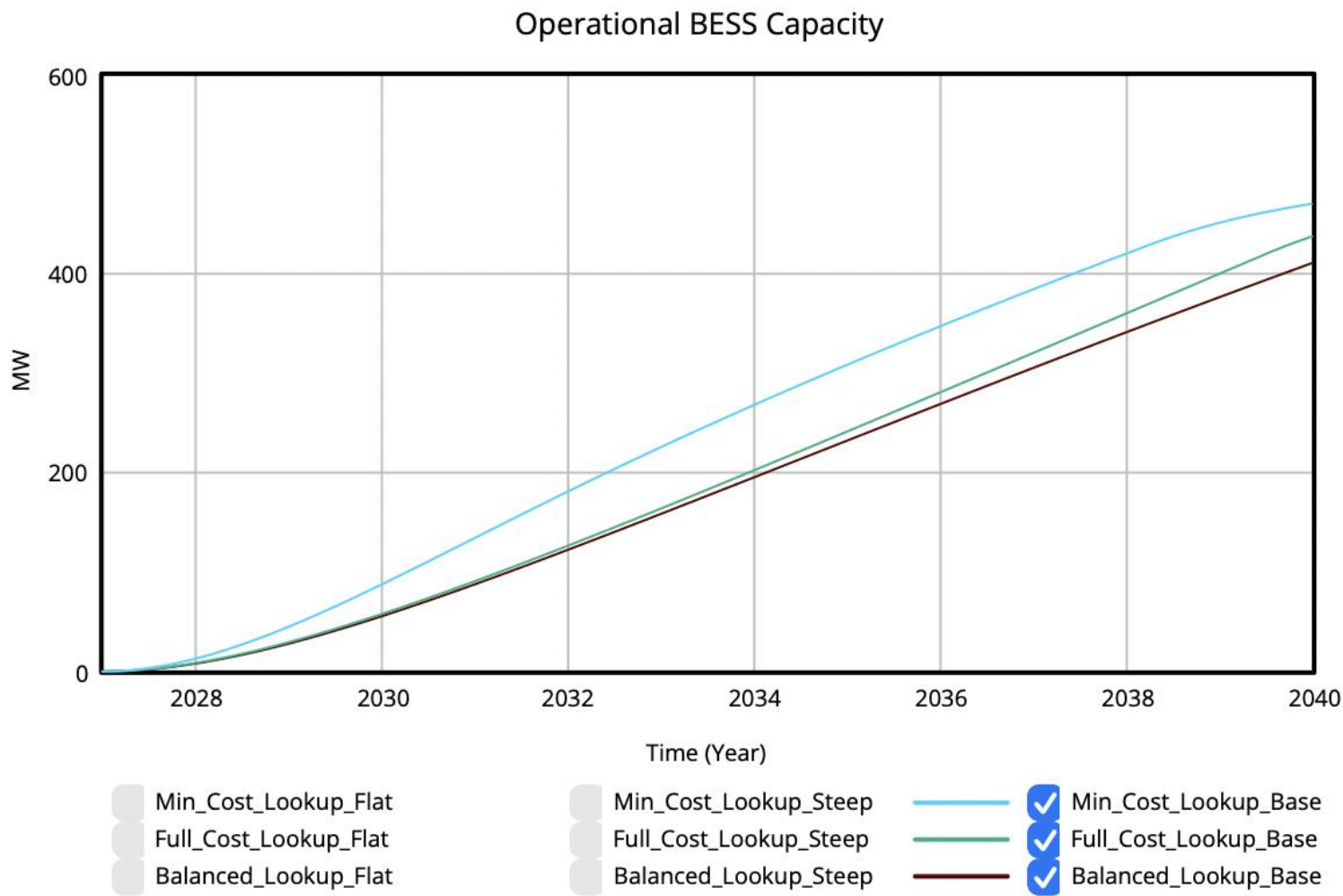
More Installed BESS Capacity → more BESS electricity generation (displaces expensive gas plant) → peak prices fall → arbitrage spread narrows → profitability falls → investment slows

B2 — Demand-side cannibalisation

More Installed BESS Capacity → more BESS charging at low prices → absorbs curtailment but raises trough prices → arbitrage spread narrows → profitability falls → investment slows

FINDING 1 — DEPLOYED BESS CAPACITY

Under the model assumptions, the scheme delivers 411 – 470 MW of 4-hour-BESS by 2039, with Minimum Cost Recovery achieving the highest deployment



411 – 470 MW (~60 MW gap)

Min Cost scenario delivers the highest deployment, followed by Full Cost scenario and Balanced Scenario.

Min Cost scenario achieves the highest Revenue-to-Cost Ratio

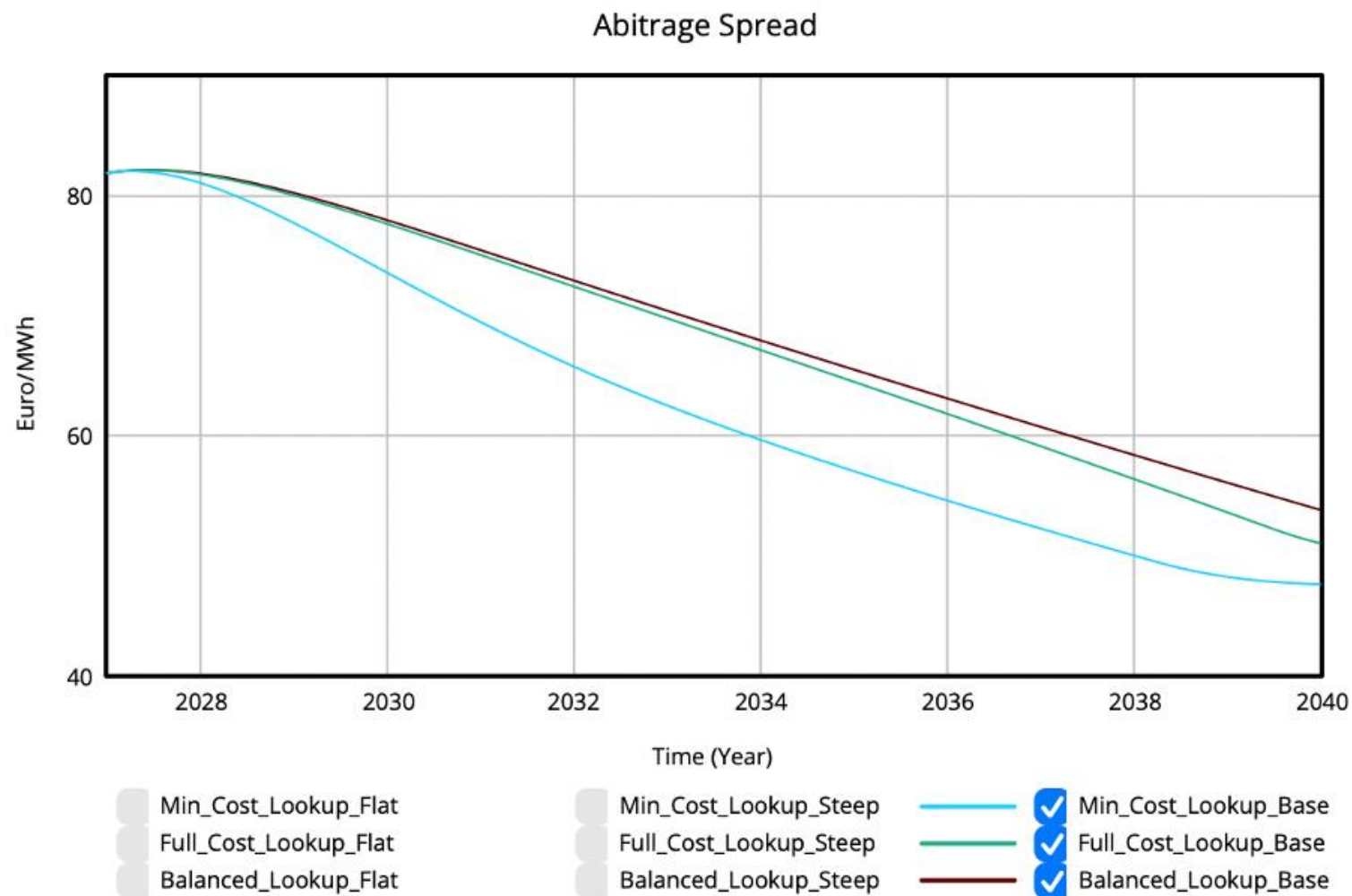
A 20% operational restriction lets developers retain 80% of revenue, keeping projects above break-even in early years.

Balanced scenario achieves the lowest Revenue-to-Cost Ratio

A 30% operational restriction reduces revenue below break-even from the outset but not low enough for the Revenue Floor (90% of LCOS) to kick in

FINDING 2 — ARBITRAGE SPREAD

As deployment is highest under Minimum Cost Recovery, the arbitrage spread falls the most, from €81.93/MWh to €47.57/MWh



**Reduction of 34% – 42% from
€81.93/MWh starting point**

The decline in arbitrage spread mirrors the deployment ranking: 1) Min Cost; 2) Full Cost; 3) Balanced.

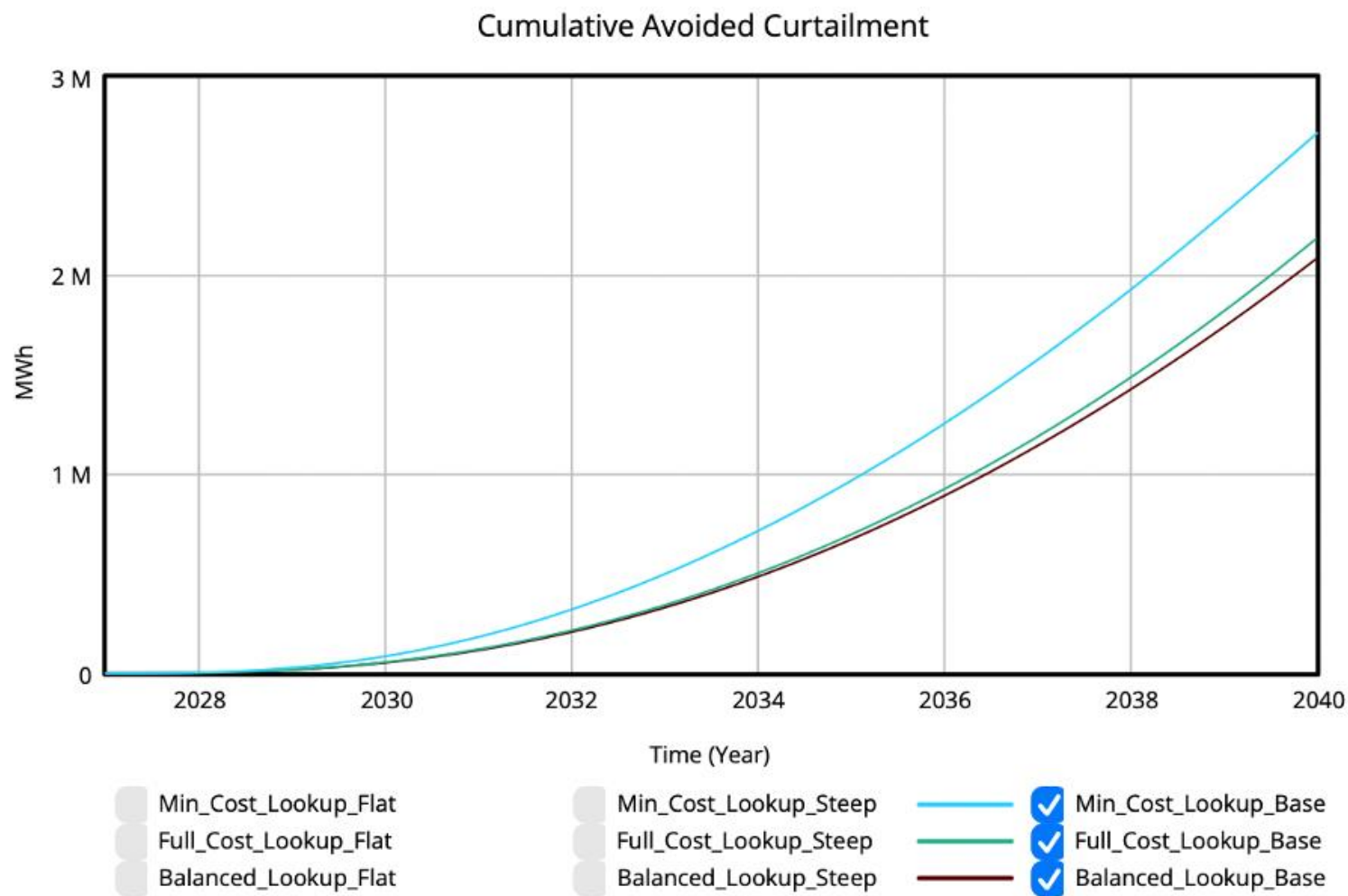
**Average Buying Price decreases in
simulation across all 3 scenarios**

The upward pressure on charging prices from greater BESS deployment is outweighed by the downward pressure of curtailment growth.

**Average Selling Price decreases in
simulation across all 3 scenarios**

As BESS capacity is deployed, generation-side price cannibalization places downward pressure on peak electricity prices

Minimum Cost Recovery avoids the most curtailment (2.71 million MWh) due to earlier and greater deployment



**2.09 million – 2.71 million MWh
(~0.62 million MWh gap)**

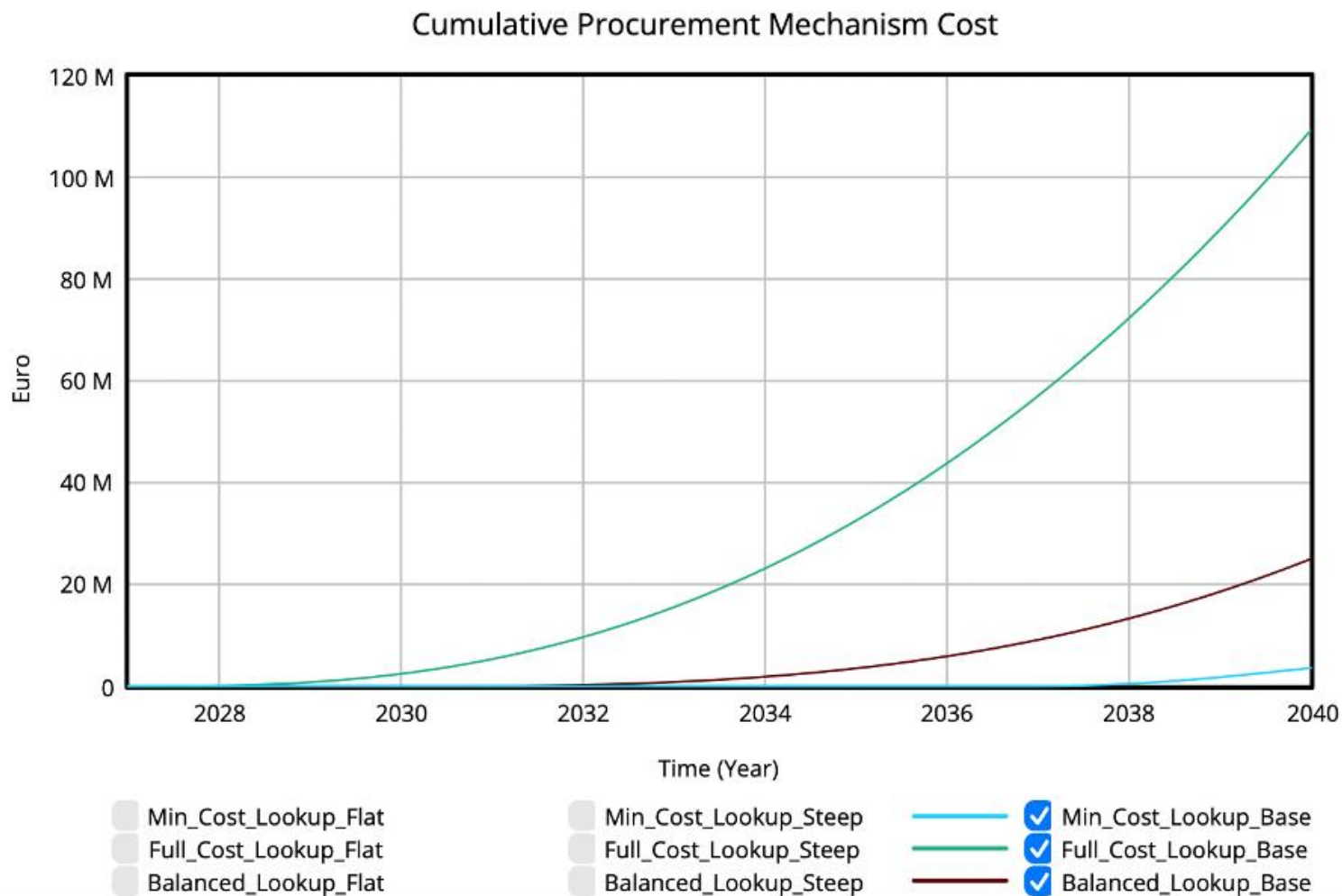
Curtailment avoided mirrors the deployment ranking: 1) Min Cost (2.71 million); 2) Full Cost (2.19 million); 3) Balanced (2.09 million).

**Earlier deployment creates a
compounding advantage**

Min Cost scenario deploys capacity earlier and reaches the highest capacity among all 3 scenarios

FINDING 4 — POLICY COST

Full Cost Recovery incurs the highest policy cost (€109.3 million) by end of 2039, around 30x that of the cheapest scenario, Minimum Cost Recovery (€3.61 million)



€3.61 million – €109.3 million by 2039

Policy cost ranking (highest to lowest): 1) Full Cost (€3.61 million); 2) Balanced (€25.04 million); 3) Min Cost (€3.61 million).

Annual Guaranteed Revenue

€4.41 million under Min Cost (RF Factor 0.7)

€5.68 million under Balanced (RF Factor 0.9)

€6.31 million under Min Cost (RF Factor 1.0)

Value for Money (Cost/ Curtailment)

€0.65 per MWh under Min Cost

€8.91 per MWh under Balanced

€41.03 per MWh under Min Cost

So what does this all mean for EirGrid (TSO)?

1 From a developer's perspective, is the Operational Envelope justified by the Revenue Floor?

- Simulation results indicate **no**: unconstrained revenues remain above break-even throughout the period, making the light-touch approach more effective
- Over a longer horizon (beyond 2039), increased price cannibalization could make the Revenue Floor more valuable risk-management tool for developers.
- An accurate LCOS is critical in determining whether revenue protection is justified.

2 Can battery projects in Ireland operate at a sufficient low cost?

- LCOS represents the average cost of storing electricity over the battery's lifetime (19 years in model).
- The simulation results are highly sensitive to the assumed LCOS (€60/MWh).
- Increasing the LCOS to €130/MWh reverses the policy ranking completely: 400.2 MW under Full Cost Recovery; 325.2 MW under Balanced; 200.1 MW under Minimal Cost Recovery

3 Contract duration may be a key determinant of scheme's attractiveness

- The model assumes a ten-year support contract with an operational lifetime of 19 years.
- If price cannibalization develops as according to the model, developers may place greater value on long-term revenue certainty beyond the initial support period (e.g. first 10 years).

LIMITATIONS

This model comes with various limitations as it represents a necessary simplification of a highly complex real-world system

■ All BESS investment occurs via procurement mechanism

The model did not quantify how much deployment would occur without the procurement mechanism or how many projects the scheme would attract relative to an unconstrained market.

■ Operational Envelope is modelled as a proportional reduction in arbitrage revenue.

The model does not capture their dispatch decision

■ Only Day-Ahead Market revenues are considered

Other important revenue streams, including the Balancing Market, DS3, and the Capacity Market are excluded from model.

■ No growth in electricity demand and all battery charge and discharge simultaneously

Likely to overstate price cannibalization, underestimate future project revenues

■ Revenue to Cost Ratio is based on annual, not lifetime revenue

■ Investment Lookup Function is judgment-based

The lookup function capture directional behaviour but not the full complexity of future electricity markets or numerical accuracy.

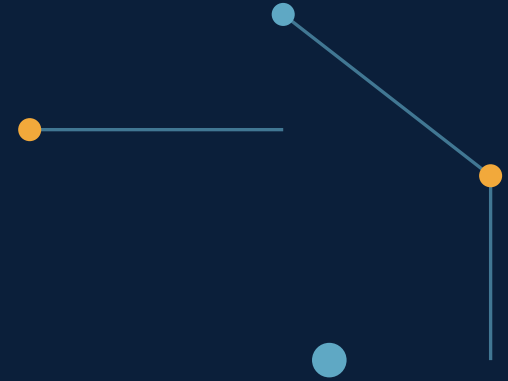
Thank You

Questions welcome

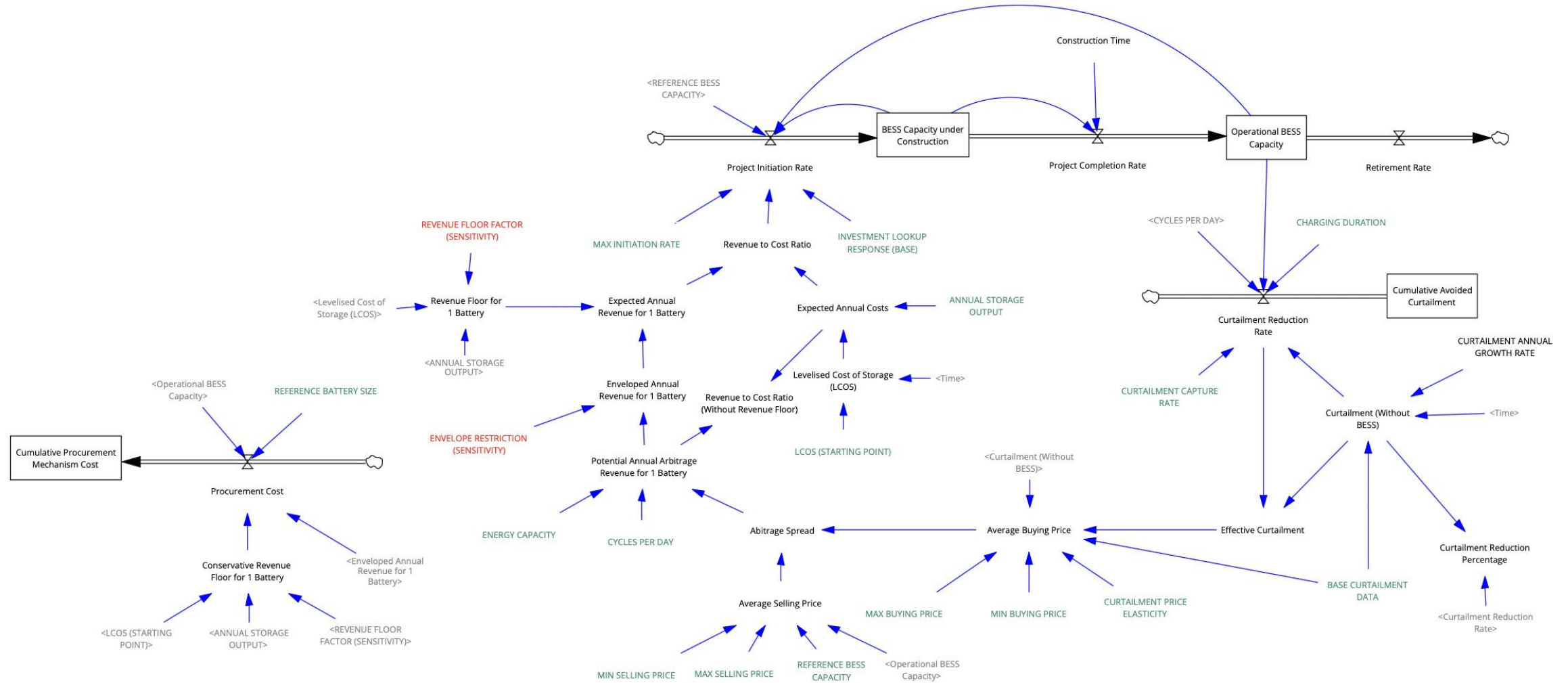
“**All models are wrong**, but some are useful”

- George E.P. Box

Appendices

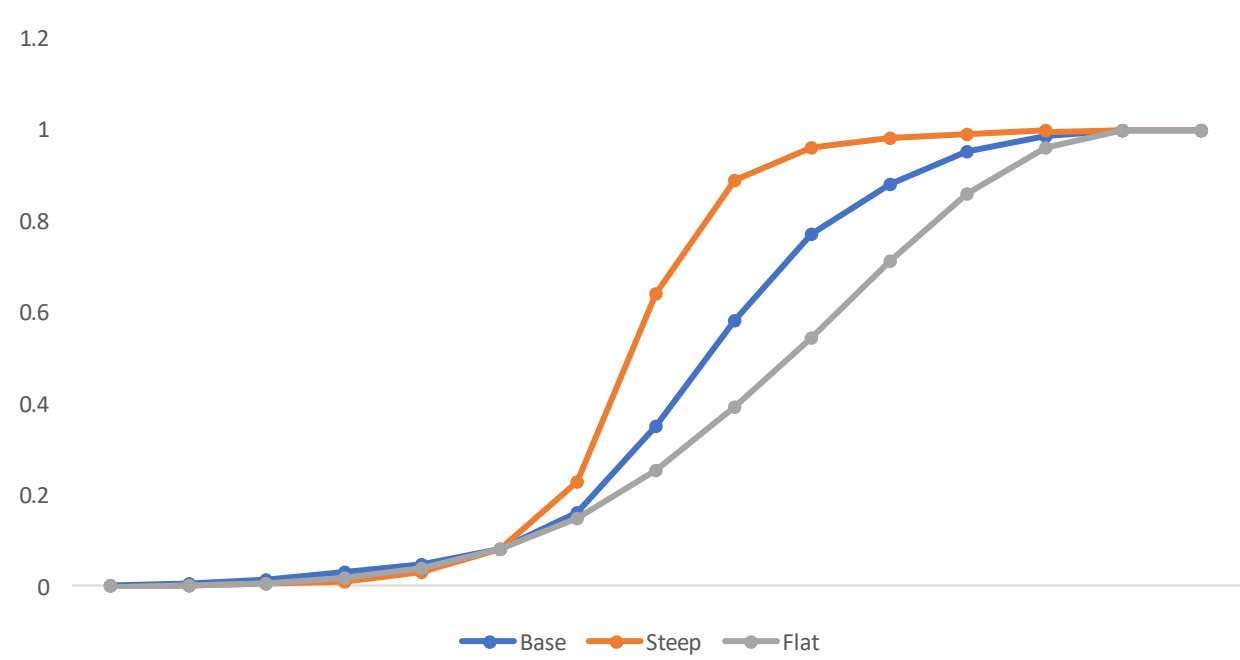


Full model structure



Complete Vensim stock-and-flow structure: 4 stocks, 4 flows, ~30 auxiliaries/constants across the curtailment, price, investment-profitability, deployment and policy-cost sub-models shown separately on Slide 4. Dense by design — not narrated live; used only if asked to walk through the full model.

Sensitivity design in full: the 9-run matrix



Steep: A highly reactive industry that waits for a distinct profitability threshold before flooding the market with capital (max increment 0.41)

Base: The central assumption used throughout the main deck

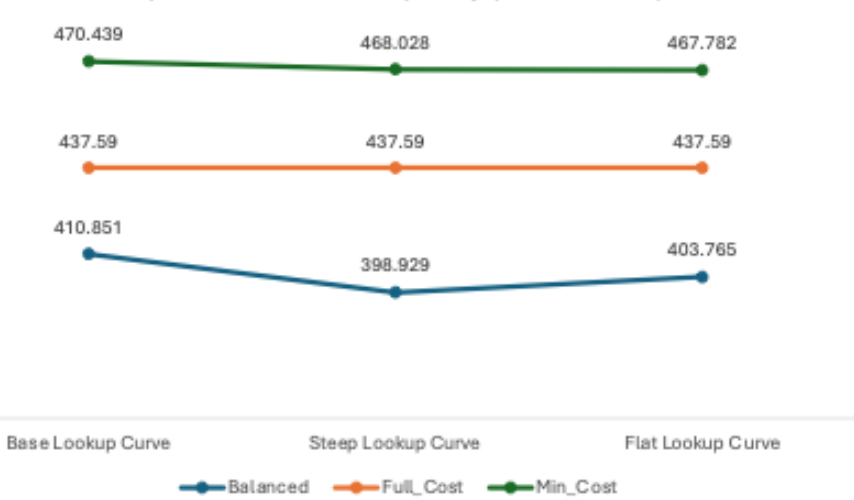
Flat: A sluggish market, heterogeneous and insensitive to rapid price signals (max increment 0.165)

	Base lookup	Steep lookup	Flat lookup
Balanced	Balanced_Lookup_Base	Balanced_Lookup_Steep	Balanced_Lookup_Flat
Maximum Cost Recovery	Full_Cost_Lookup_Base	Full_Cost_Lookup_Steep	Full_Cost_Lookup_Flat
Minimum Cost Recovery	Min_Cost_Lookup_Base	Min_Cost_Lookup_Steep	Min_Cost_Lookup_Flat

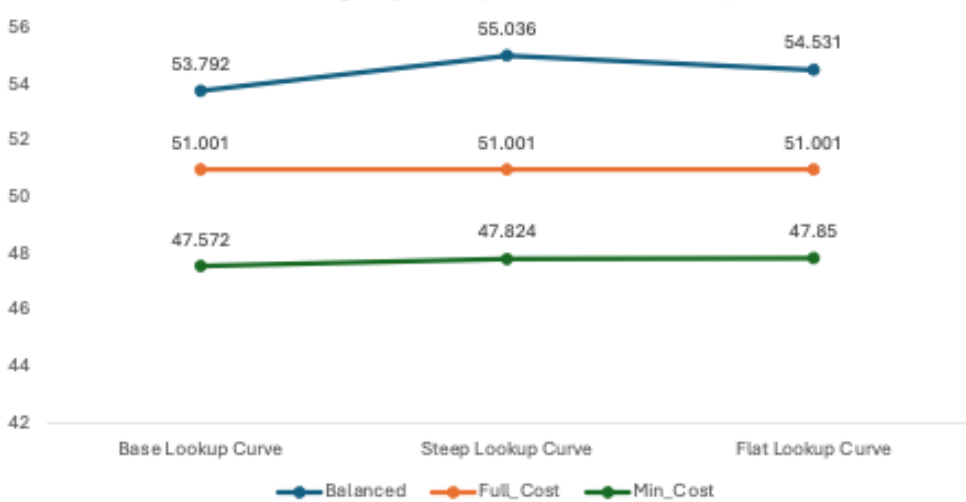
SENSITIVITY ANALYSIS RESULTS: BEHAVIOUR MODE SENSITIVITY

Across all four output variables, the ranking remains unchanged at end 2039 under the three different lookup curves.

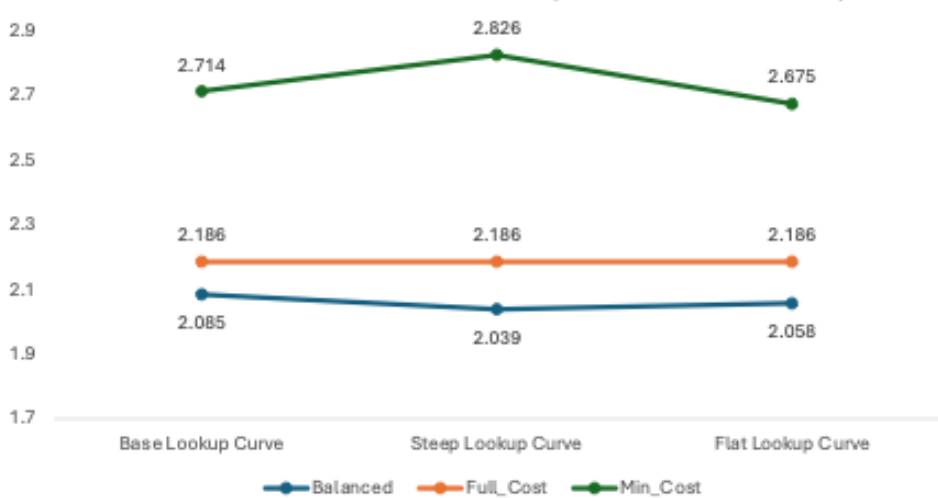
Operational BESS Capacity (MW at 2039)



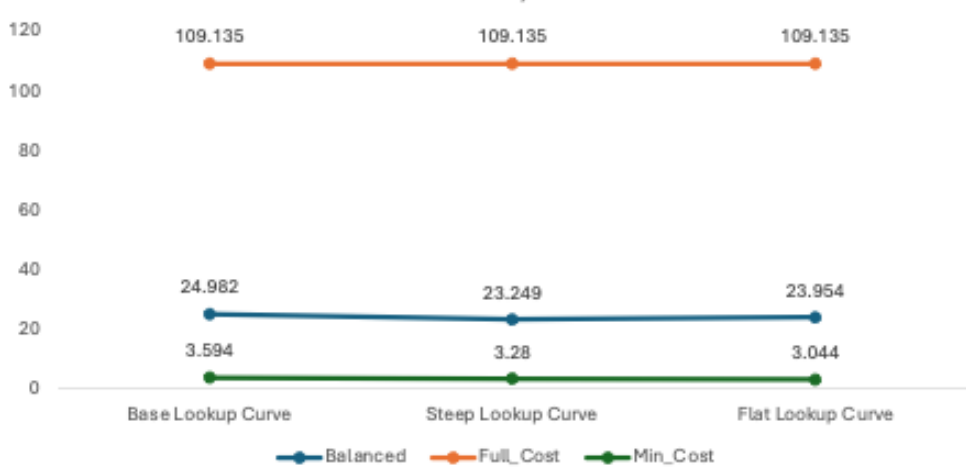
Arbitrage Spread (EUR/MWh at 2039)



Cumulative Avoided Curtailment (MWh, million at 2039)



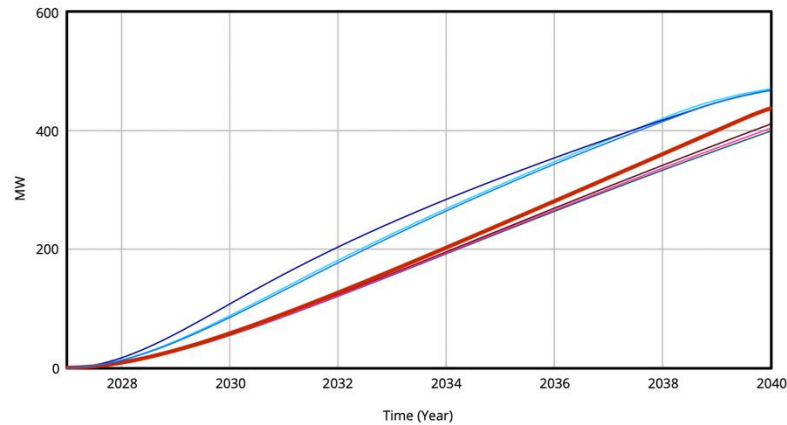
Cumulative Procurement Mechanism Cost (EUR, million at 2039)



SENSITIVITY ANALYSIS RESULTS: POLICY SENSITIVITY

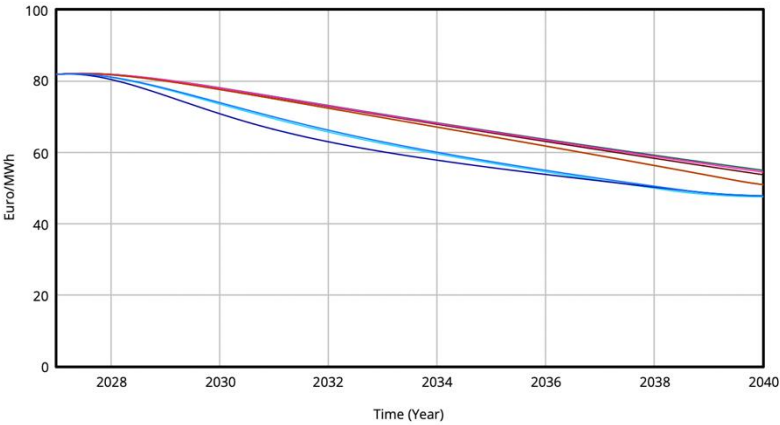
Across all simulation runs, no output exhibits oscillation, runaway growth, or change in behaviour mode

Operational BESS Capacity



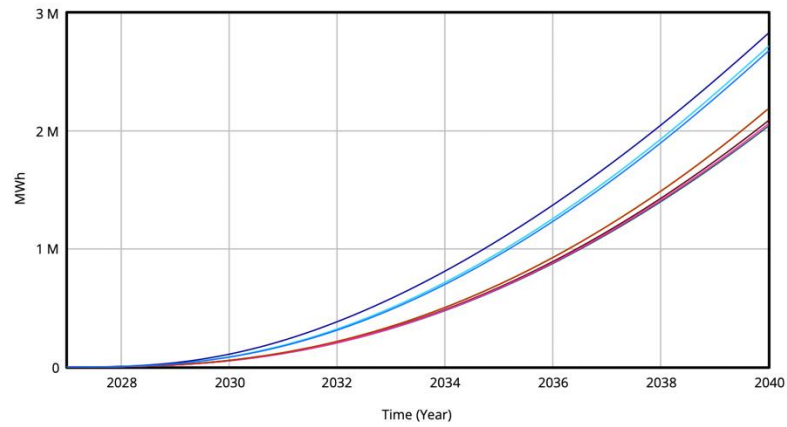
- Min_Cost_Lookup_Flat
- Min_Cost_Lookup_Steep
- Min_Cost_Lookup_Base
- Full_Cost_Lookup_Flat
- Full_Cost_Lookup_Steep
- Full_Cost_Lookup_Base
- Balanced_Lookup_Flat
- Balanced_Lookup_Steep
- Balanced_Lookup_Base

Abitrage Spread



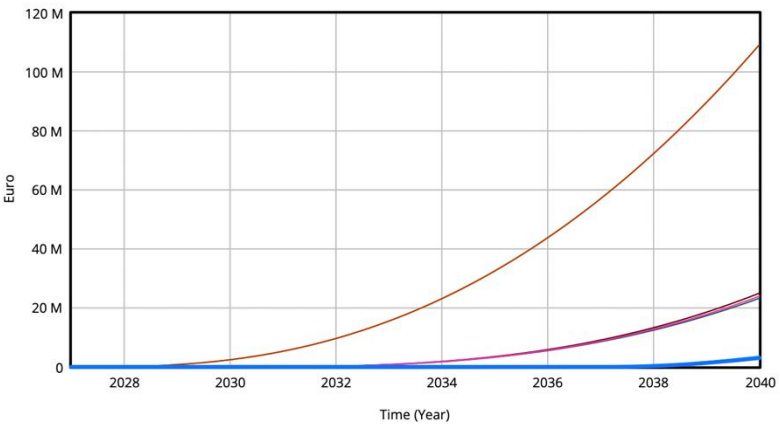
- Min_Cost_Lookup_Flat
- Min_Cost_Lookup_Steep
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- Full_Cost_Lookup_Flat
- Full_Cost_Lookup_Steep
- Full_Cost_Lookup_Base
- Balanced_Lookup_Flat
- Balanced_Lookup_Steep
- Balanced_Lookup_Base

Cumulative Avoided Curtailment



- Min_Cost_Lookup_Flat
- Min_Cost_Lookup_Steep
- Min_Cost_Lookup_Base
- Full_Cost_Lookup_Flat
- Full_Cost_Lookup_Steep
- Full_Cost_Lookup_Base
- Balanced_Lookup_Flat
- Balanced_Lookup_Steep
- Balanced_Lookup_Base

Cumulative Procurement Mechanism Cost



- Min_Cost_Lookup_Flat
- Min_Cost_Lookup_Steep
- Min_Cost_Lookup_Base
- Full_Cost_Lookup_Flat
- Full_Cost_Lookup_Steep
- Full_Cost_Lookup_Base
- Balanced_Lookup_Flat
- Balanced_Lookup_Steep
- Balanced_Lookup_Base

Stock variables represent quantities that accumulate over time through the net effect of inflows and outflows

Stock Variable	Equation	Unit	Description
BESS Capacity under Construction	$\frac{dBCS}{dt} = PIR - PCR$ Where: $BCS = \text{BESS Capacity under Construction}$$\frac{dBCS}{dt} = \text{Rate of Change of BCS}$$PIR = \text{Project Initiation Rate}$$PCR = \text{Project Completion Rate}$	MW	Total BESS capacity that has entered the development pipeline but has not yet become operational. The stock increases through project initiation and decreases as projects complete construction. Initial value = 0 MW.
Operational BESS Capacity	$\frac{dK}{dt} = PCR - RR$ Where: $K = \text{Operational BESS Capacity}$$\frac{dK}{dt} = \text{Rate of Change of K}$$PCR = \text{Project Completion Rate}$$RR = \text{Retirement Rate}$	MW	Total BESS capacity that has completed construction and entered commercial operation. This stock determines arbitrage revenues, curtailment absorption, and procurement costs. Initial value = 0 MW.
Cumulative Avoided Curtailment	$\frac{dCAR}{dt} = CRR$ Where: $CAR = \text{Cumulative Avoided Curtailment}$$\frac{dCAR}{dt} = \text{Rate of Change of CAR}$$CRR = \text{Curtailment Reduction Rate}$	MWh	Total renewable energy curtailment absorbed by the BESS fleet since the start of the simulation. Initial value = 0 MWh.
Cumulative Procurement Mechanism Cost	$\frac{dCPMC}{dt} = PROC$ Where: $CPMC = \text{Cumulative Procurement Mechanism Cost}$$\frac{dCPMC}{dt} = \text{Rate of Change of CPMC}$$PROC = \text{Procurement Cost}$	Euro	Total of the net cost of the procurement mechanism since 2027. Positive values indicate a net fiscal cost, while negative values indicate net revenue. Initial value = 0 Euro.

Flow variables represent the rates at which stocks change, governing their accumulation or depletion over time (1/2)

Stock Variable	Equation	Unit	Description
Project Initiation Rate	$PIR = \begin{cases} 0, & BCS + K \geq K_{ref} \\ I_{max}f_{lookup}(RC), & BCS + K < K_{ref} \end{cases}$ Where: PIR = Project Initiation Rate BCS = BESS Capacity under Construction K = Operational BESS Capacity K_{ref} = REFERENCE BESS CAPACITY I_{max} = MAX INITIATION RATE f_{lookup} = INVESTMENT LOOKUP RESPONSE $RC_{With Floor}$ = Revenue to Cost Ratio	MW/Year	New BESS capacity entering construction each year. Investment activity ceases once the combined operational and under-construction capacity reaches the procurement cap of 500 MW.
Project Completion Rate	$PCR = \frac{BCS}{T_{Construction}}$ Where: PCR = Project Completion Rate BCS = BESS Capacity under Construction $T_{Construction}$ = Construction Time	MW/Year	Rate at which projects complete construction and becomes operational. The formulation represents a first-order delay with an average construction period of two years.
Retirement Rate	$RR = 0$ Where: RR = Retirement Rate	MW/Year	BESS retirements are not represented within the 2027–2039 simulation horizon, as the assumed battery lifetime of 19 years exceeds the model period. Consequently, all deployed BESS assets remain operational throughout the simulation and no capacity is retired.
Curtailment Reduction Rate	$CRR = \min(C_{Without BESS}, Km365c)$ Where: CRR = Curtailment Reduction Rate $C_{Without BESS}$ = Curtailment (Without BESS) K = Operational BESS Capacity h = CHARGING DURATION n = CYCLES PER DAY c = CURTAILMENT CAPTURE RATE	Percentage	Annual renewable curtailment absorbed by the operational BESS fleet. Curtailment reduction is calculated as the smaller of total available curtailment and the fleet's curtailment absorption capacity. In other words, the model assumes that batteries cannot absorb more curtailment than exists, nor can they absorb more than they are physically capable of absorbing.

Flow variables represent the rates at which stocks change, governing their accumulation or depletion over time (2/2)

Stock Variable	Equation	Unit	Description
Procurement Cost	$PROC = \max(0, RF_{Conservative} - AR_{Enveloped}) (\frac{K}{S_{ref}})$ Where: $PROC = \text{Procurement Cost}$$RF_{Consevative} = \text{Conservative Revenue Floor for 1 Battery}$$AR_{Enveloped} = \text{Enveloped Annual Revenue for 1 Battery}$$K = \text{Operational BESS Capacity}$$S_{ref} = \text{REFERENCE BATTERY SIZE}$	Euro/Year	Total procurement expenditure required to guarantee the Conservative Revenue Floor. The cost increases when battery revenues fall below the revenue floor and scales with the amount of operational BESS capacity.

Auxiliary variables combine model assumptions and system conditions to calculate key outcomes. This table focuses on the **curtailment** sub-model

Auxiliary Variable	Equation	Unit	Description
Curtailment Without BESS	$C_{Without\ BESS} = C_{2027}(1 + g)^{(t-2027)}$ Where: $C_{Without\ BESS} = \text{Curtailment (Without BESS)}$ $C_{2027} = \text{BASE CURTAILMENT DATA}$ $g = \text{CURTAILMENT GROWTH RATE}$ $t = \text{Time}$	MWh/Year	Estimates annual curtailment assuming no new BESS capacity is deployed. (EirGrid, 2025a) and author’s calculation. See CURTAILMENT ANNUAL GROWTH RATE subsection under Constants above.
Effective Curtailment	$C_E = \max(0, C_{Without\ BESS} - CRR)$ Where: $C_E = \text{Effective Curtailment}$ $C_{Without\ BESS} = \text{Curtailment (Without BESS)}$ $CRR = \text{Curtailment Reduction Rate}$	MWh	Estimates the annual curtailment remaining after absorption by the operational BESS fleet. The value is constrained to a minimum of zero.
Curtailment Reduction Percentage	$CRP = \frac{CRR}{C_{Without\ BESS}}$ Where: $CRP = \text{Curtailment Reduction Percentage}$ $CRR = \text{Curtailment Reduction Rate}$ $C_{Without\ BESS} = \text{Curtailment (Without BESS)}$	Percentage	Share of projected curtailment absorbed by the operational BESS fleet.

Auxiliary variables combine model assumptions and system conditions to calculate key outcomes. This table focuses on the **electricity price** sub-model (1/2)

Auxiliary Variable	Equation	Unit	Description
Average Buying Price	$P_b = \left[P_{b,min} + (P_{b,max} - P_{b,min}) \left(1 - \frac{C_E}{C_{Without\ BESS}} \right) \right] \left(\frac{C_B}{C_{2027}} \right)^\varepsilon$ Where: $P_b = \text{Average Buying Price}$ $P_{b,min} = \text{MIN BUYING PRICE}$ $P_{b,max} = \text{MAX BUYING PRICE}$ $C_E = \text{Effective Curtailment}$ $C_{Without\ BESS} = \text{Curtailment (Without BESS)}$ $C_{2027} = \text{BASE CURTAILMENT DATA}$ $\varepsilon = \text{CURTAILMENT PRICE ELASTICITY}$	Euro/MWh	Average electricity price paid during battery charging periods. The charging price increases as curtailment levels decline, reflecting the assumption that higher curtailment (greater surplus generation) suppresses electricity prices, while lower curtailment leads to higher market prices.
Average Selling Price	$P_s = \max[P_{s,min}, P_{s,max} - (P_{s,max} - P_{s,min}) \frac{K}{K_{ref}}]$ Where: $P_s = \text{Average Selling Price}$ $P_{s,min} = \text{MIN SELLING PRICE}$ $P_{s,max} = \text{MAX SELLING PRICE}$ $K = \text{Operational BESS Capacity}$ $K_{ref} = \text{REFERENCE BESS CAPACITY}$	Euro/MWh	Average electricity price received by battery operators during discharge periods when stored electricity is sold back to the grid. The equation assumes that the Average Selling Price will decline as operational BESS capacity increases. A price floor is imposed through the max function, which ensures that the Average Selling Price cannot fall below the MIN SELLING PRICE. Limitations Electricity demand is assumed to remain constant throughout the simulation period and is therefore excluded from the price formulation. Any change in selling prices is assumed to be caused only by changes in BESS deployment. This represents an important limitation, as electricity demand is a major determinant of market prices. For example, if electricity demand increases faster than the available BESS discharge capacity, selling prices may rise despite additional BESS deployment. Under such conditions, the Minimum Selling Price would likely increase over time. The current model is unable to capture this behaviour because demand dynamics are not represented. As a result, the assumption of a fixed Minimum Selling Price should be interpreted as a modelling simplification rather than a realistic representation of future market conditions.

Auxiliary variables combine model assumptions and system conditions to calculate key outcomes. This table focuses on the **electricity price** sub-model (2/2)

Auxiliary Variable	Equation	Unit	Description
Arbitrage Spread	$S = P_s - P_b$ Where: $S = \text{Arbitrage Spread}$ $P_s = \text{Average Selling Price}$ $P_b = \text{Average Buying Price}$	Euro/MWh	Difference between average selling and buying prices.

Auxiliary variables combine model assumptions and system conditions to calculate key outcomes. This table focuses on the investment profitability sub-model (1/3)

Auxiliary Variable	Equation	Unit	Description
Levelised Cost of Storage (LCOS)	$LCOS_t = LCOS_{2027}(1 - 0.02736)^{(t-2027)}$ Where: $LCOS_t = \text{Levelised Cost of Storage (LCOS)}$ $LCOS_{2027} = \text{LCOS (STARTING POINT)}$ $t = \text{Time}$	Euro/MWh	Dynamic LCOS that reflects declining capital expenditure (CAPEX) and increasing operating expenditure (OPEX) over time, based on cost projections reported by KPMG. (KPMG, 2023) projects CAPEX values of €354/kWh in 2020, €248/kWh in 2030, and €150/kWh in 2050. These projections imply a compound annual growth rate (CAGR) of -3.5% between 2020 and 2030, and -2.5% between 2030 and 2050. As the model baseline year is 2027, the annual CAPEX reduction rate of -3.5% is applied from 2027 to 2030. From 2030 onwards, the annual reduction rate of -2.5% is applied until the end of the simulation period in 2039. When the entire modelling period from 2027 to 2039 is considered, the implied average annual cost reduction rate is approximately -2.7%. This value is therefore used to represent the long-term decline in LCOS within the model. Limitation Applying the CAPEX reduction rate directly to LCOS therefore assumes that changes in capital costs are representative of changes in overall lifecycle costs.
Expected Annual Costs	$EC = LCOS_t O$ Where: $EC = \text{Expected Annual Costs}$ $LCOS_t = \text{Levelised Cost of Storage}$ $O = \text{ANNUAL STORAGE OUTPUT}$	Euro/Year	Annualised cost of operating a representative battery project.
Potential Annual Arbitrage Revenue for 1 Battery	$AR_{potential} = En365S$ Where: $AR_{potential} = \text{Potential Annual Arbitrage Revenue}$ $E = \text{ENERGY CAPACITY}$ $n = \text{CYCLES PER DAY}$ $S = \text{Arbitrage Spread}$	Euro/Year	Gross annual arbitrage revenue before operational constraints are applied. The model assumes that participation in the day-ahead electricity market is the sole source of market-based revenue for battery operators. No additional revenue streams, such as ancillary services or capacity market payments, are included. Consequently, battery revenues are derived exclusively from energy arbitrage and the Revenue Floor mechanism.

Auxiliary variables combine model assumptions and system conditions to calculate key outcomes. This table focuses on the investment profitability sub-model (2/3)

Auxiliary Variable	Equation	Unit	Description
Revenue to Cost Ratio (Without Revenue Floor)	$RC_{No\ Floor} = \frac{AR}{EC}$ Where: $RC_{No\ Floor} = \text{Revenue to Cost Ratio (Without Revenue Floor)}$ $AR = \text{Potential Annual Arbitrage Revenue}$ $EC = \text{Expected Annual Costs}$	Dimensionless	Project profitability calculated using the revenues and costs incurred within a single year. Limitation This measure is not a Net Present Value (NPV) calculation. Revenue is based solely on the projected cash flows generated within the given year. Future revenues and costs are not discounted to their present value. The metric should therefore be interpreted as an annual profitability indicator rather than a measure of lifetime project value.
Enveloped Annual Revenue for 1 Battery	$AR_{Enveloped} = AR(1 - \rho)$ Where: $AR_{Enveloped} = \text{Enveloped Annual Revenue for 1 Battery}$ $AR = \text{Potential Annual Arbitrage Revenue}$ $\rho = \text{ENVELOPE RESTRICTION (SENSITIVITY)}$	Euro/Year	Annual revenue after operational restrictions imposed by the operational envelope.
Expected Annual Revenue for 1 Battery	$AR_{With\ Floor} = \max(F, AR_{Enveloped})$ Where: $AR_{With\ Floor} = \text{Expected Annual Revenue for 1 Battery}$ $F = \text{Revenue Floor for 1 Battery}$ $AR_{Enveloped} = \text{Enveloped Annual Revenue}$	Euro/Year	The annual revenue expected from a single battery after applying the revenue floor mechanism.
Revenue Floor for 1 Battery	$RF_t = LCOS_t O \phi$ Where: $RF_t = \text{Revenue Floor for 1 Battery}$ $LCOS_t = \text{Levelised Cost of Storage}$ $O = \text{ANNUAL STORAGE OUTPUT}$ $\phi = \text{REVENUE FLOOR FACTOR}$	Euro/Year	Minimum annual revenue guaranteed to eligible projects under the procurement mechanism.
Revenue to Cost Ratio	$RC_{With\ Floor} = \frac{ER}{EC}$ Where: $RC_{With\ Floor} = \text{Revenue to Cost Ratio}$ $ER = \text{Expected Annual Revenue}$ $EC = \text{Expected Annual Cost}$	Dimensionless	Similar to the Revenue-to-Cost Ratio (without Revenue Floor), but in this case annual revenue includes the Revenue Floor mechanism. The same limitation identified for the Revenue-to-Cost Ratio (without Revenue Floor) also applies to this metric. It remains a single-year measure and does not account for discounted lifetime cash flows.

Auxiliary variables combine model assumptions and system conditions to calculate key outcomes. This table focuses on the **investment profitability** sub-model (3/3)

Auxiliary Variable	Equation	Unit	Description
INVESTMENT LOOKUP RESPONSE	$I = f_{lookup}(RC)$ Where: $I = INVESTMENT\ LOOKUP\ RESPONSE$ $f_{lookup} = Lookup\ Table$ $RC = Revenue\ to\ Cost\ Ratio$	Dimensionless	Converts project profitability into a proportion of the maximum annual project initiation rate. Refer to the subsection below for the steps used to define a non-linear table function, as outlined by (Stermann, 2000).

Auxiliary variables combine model assumptions and system conditions to calculate key outcomes. This table focuses on the **policy cost** sub-model

Auxiliary Variable	Equation	Unit	Description
Conservative Revenue Floor for 1 Battery	$RF_{Conservative} = LCOS_{2027} O \phi$ Where: $RF_{Conservative} = \text{Conservative Revenue Floor}$ $LCOS_{2027} = LCOS \text{ (STARTING POINT)}$ $O = \text{ANNUAL STORAGE OUTPUT}$ $\phi = \text{REVENUE FLOOR FACTOR}$	Euro/Year	Annual Revenue Floor calculated using the initial LCOS value, which remains fixed throughout the simulation period. This approach is used to produce a conservative estimate of policy cost. Limitation The Revenue Floor is calculated using only the baseline LCOS and does not account for the projected decline in LCOS over time. As a result, future reductions in battery costs are not reflected in the calculation of policy support payments. This assumption is likely to overestimate the required level of support and therefore produces a conservative estimate of policy cost. In practice, declining LCOS would reduce the revenue support required from the mechanism, resulting in lower policy costs than those projected by the model.

The complete list of model constants are as below (1/3)

Constant	Value	Unit	Description	Source
ANNUAL STORAGE OUPUT	105,120	MWh/Year	Annual energy delivered by a 100 MW / 400 MWh battery cycling at 0.863 times per day for 365 days. $ANNUAL\ STORAGE\ OUTPUT$ $= Utilisation \times Efficiency \times Power \times 365$ $= 0.8 \times 0.9 \times 400 \times 365$ $= 105,120\ MWh$	Author’s own calculation based on utilisation and efficiency assumptions from (Rangelova & Jones, 2025).
BASE CURTAILMENT DATA	258,000	MWh/Year	Projected renewable energy curtailment in Ireland for 2027, estimated from EirGrid’s ECP 2.4 scenario modelling using direct-ratio scaling of the reported curtailment percentage. $Curtailment_{2027} = \frac{Curtailment\ (\%)}{Constraint\ (\%)} \times Constraint\ (TWh)$ $= \frac{1.4}{9} \times 1.66$ $= 0.258\ TWh$ $= 258,000\ MWh$	(EirGrid, 2025a)
CHARGING DURATION	4	Hours/Cycle	Fixed battery duration. One full cycle consists of 4 hours charging at rated power.	Model definition
CONSTRUCTION TIME	2	Year	Average time from contract award to commercial operation for a utility-scale BESS project.	Author assumption
CURTAILMENT ANNUAL GROWTH RATE	0.28	Dimensionless	Annual growth rate applied to baseline of curtailment volumes. Partially derived from EirGrid ECP 2.4 scenario modelling.	See subsection below.
CURTAILMENT CAPTURE RATE	0.75	Dimensionless	Fraction of battery capacity that successfully absorbs curtailment per cycle.	Author assumption
CURTAILMENT PRICE ELASTICITY	0.0626	Dimensionless	Governs how much the average buying price rises as curtailment is absorbed.	(Swinand & O’Mahoney, 2015) See subsection below.
CYCLES PER DAY	0.8	Dimensionless	Cycles per day is set to 0.8 and is dimensionless. In energy storage systems, this is commonly referred to as round-trip efficiency, which measures the share of energy retained after a full charge–discharge cycle. It is calculated as the product of charging efficiency (the proportion of electricity stored during charging) and discharging efficiency (the proportion of stored electricity that is recovered during discharge). In Ember’s levelised cost of storage (LCOS) methodology, a value of 0.8 is used to represent this efficiency (Rangelova & Jones, 2025).	(Rangelova & Jones, 2025)

The complete list of model constants are as below (2/3)

Constant	Value	Unit	Description	Source
ENERGY CAPACITY	360	MWh	Total energy storage of a 4-hour battery unit. $100\text{ MW} \times 4\text{ hour} \times 90\%\text{ efficiency} = 400\text{ MWh}$	Author’s calculation.
LCOS (STARTING POINT)	67	Euro/MWh	Levelised Cost of Storage in the base year (2027) using assumptions from (EirGrid, 2023b; KPMG, 2023; Lazard, 2025; Rangelova & Jones, 2025).	(EirGrid, 2023b; KPMG, 2023; Lazard, 2025; Rangelova & Jones, 2025) See subsection below.
MAX BUYING PRICE	103.76	Euro/MWh	Electricity price at which the BESS purchases power when the full 500 MW of storage capacity is deployed. In this model, all units are assumed to charge and discharge at the same time each day. Each unit is assumed to complete one full charge–discharge cycle per day. In summary, this value is calculated as the average of the four highest-priced hours within the off-peak period each day over the period 2023 to 2025	(Ember, n.d.) See subsection below.
MIN BUYING PRICE	81.05	Euro/MWh	Baseline charging cost for a battery asset before any procured battery capacity enter the system. The value is calculated as the average of the four lowest-priced hours each day over the period 2023 to 2025.	(Ember, n.d.) See subsection below.
MAX SELLING PRICE	162.98	Euro/MWh	Peak electricity selling price before any cannibalisation effect from procured BESS capacity. The value is calculated as the average of the four highest-priced hours each day over the period 2023 to 2025.	(Ember, n.d.) See subsection below.
MIN SELLING PRICE	112.07	Euro/MWh	Electricity price at which the BESS sells when the full 500 MW of procured BESS capacity has entered the market , representing a state of full market saturation. This value is calculated as the 10 th percentile of the distribution of the four highest-priced hours for each day over the period of 2023 to 2025. The author acknowledges that this threshold involves significant limitations and an element of arbitrariness in the choice of percentile (for example, why 10th percentile rather than 5th). This is discussed further in the subsection below.	(Ember, n.d.) See subsection below.
EIRGRID’S SHARE	0.30	Dimensionless	EirGrid’s portion of revenue above the floor under the 70/30 revenue-sharing arrangement specified in the LDES Procurement Mechanism.	(EirGrid, 2025c)
MAX INITIATION RATE	500	MW/Year	Maximum annual new BESS capacity that the market can bring online, constrained by supply chain capacity, permitting timelines, and grid connection limits, as well as the maximum procurement volume set out by EirGrid in the consultation paper.	(EirGrid, 2025c) and author’s assumptions.
REFERENCE BATTERY SIZE	100	MW	Power capacity of the reference battery unit used for per-project revenue and cost calculations. Do not confuse with REFERENCE BESS CAPACITY (the total fleet size).	Model definition.

The complete list of model constants are as below (3/3)

Constant	Value	Unit	Description	Source
REFERENCE BESS CAPACITY	500	MW	Total BESS fleet size at full build-out, used as the denominator in cannibalisation and procurement cost scaling equations.	Model definition.
ENVELOPE RESTRICTION (SENSITIVITY)	0.3; 0.7; 0.2	Dimensionless	The fraction of battery capacity over which EirGrid retains dispatch control under the Operational Envelope. When this restriction is applied, the operator loses the ability to optimise that portion of the battery's capacity for arbitrage. Revenue is reduced proportionally: a restriction of 0.30 reduces arbitrage revenue by 30%, a restriction of 0.70 reduces it by 70%. Three values are used across the policy scenarios: 0.30 (Base Case), 0.70 (Full Cost Recovery), and 0.20 (Minimal Cost Recovery). See Table X for the full scenario design. Limitation In EirGrid's consultation document, the operational envelope is defined in terms of operating hours and constraints on charging and discharging behaviour, not as a direct revenue reduction. This model simplifies that mechanism by treating the restriction as a proportional reduction in arbitrage revenue, avoiding the need to simulate intraday dispatch decisions explicitly. This may overstate the revenue impact at lower restriction levels, where operators with greater dispatch freedom could partially compensate through strategic timing.	(EirGrid, 2025c)
REVENUE FLOOR FACTOR (SENSITIVITY)	0.9; 0.7; 1.0	Dimensionless	The fraction of LCOS guaranteed to operators as a minimum annual revenue under the procurement mechanism. A value of 1.0 represents full cost recovery; a value of 0.0 implies no floor.	(EirGrid, 2025c)